

- 9 (a) State what is meant by the mass defect of a nucleus.

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..... [2]

- (b) The nuclear fusion reaction for the formation of helium-4 from deuterium is represented by

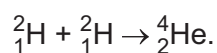


Table 9.1 shows the masses of the nuclides involved in this reaction.

Table 9.1

nuclide	nuclide mass/u
${}^2_1\text{H}$	2.013553
${}^4_2\text{He}$	4.001505

Calculate the energy released in the formation of 1.00 mol of helium-4.

energy = J [4]

